SIMATS ENGINEERING
ASSESSMENT TOOL 6

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**
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COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3 **Duration: 1 Hour**
Total Marks:30

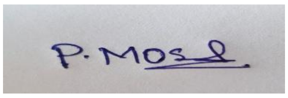
Annexure A

sDry Run Evaluation

Code of Conduct:

I P.Moshiya(192572142) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:

```
        A
       / \
      B   C
     /\   /\
    D  E F  G
```

Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	**Queue**	**Visited Node**
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,

```
        A
       / \
      B   C
     /\   /\
    D  E F  G
```

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**
   1 3 6
   5 0 2
   4 7 8

Goal State :
1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1			
2			
3			
4			
5			

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CO2 – ASSESSMENT TOOL 1

Application Based Questions

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

QUESTIONS: 3 **Duration:** 1 Hour **Total Marks:** 30

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Total Marks (30)
Understanding of Problem Context	20% (2)	/2	/2	/2	/6
Application of Concepts	30% (3)	/3	/3	/3	/9
Problem-Solving Approach	20% (2)	/2	/2	/2	/6
Accuracy of Solution	20% (2)	/2	/2	/2	/6
Clarity	10% (1)	/1	/1	/1	/3
Total per Question	10 Marks	/10	/10	/10	/30

100

1. Breadth-First Search (BFS)

Algorithm

1. Start from the root node.
2. Mark the node as visited.
3. Insert it into the queue.
4. While the queue is not empty:
   - Remove the front node.
   - Visit the node.
   - Add all its unvisited adjacent nodes to the queue.
5. Repeat until the queue becomes empty.

Graph

```
     A
   /   \
  B     C
 / \   / \
D   E F   G
```

Dry Run Table

Iteration	Queue	Visited Node
1	B, C	A

Iteration	Queue	Visited Node
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

Answers

1. BFS Traversal Order

A → B → C → D → E → F → G

2. Queue Contents After Each Iteration

Iteration	Queue
1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

2. Depth-First Search (DFS)

Algorithm

1. Start from the root node.
2. Visit the node and mark it as visited.
3. Move to the leftmost unvisited child.
4. Continue until no child exists.
5. Backtrack to the previous node.
6. Visit remaining unvisited nodes.
7. Repeat until all nodes are visited.

Dry Run Table

Iteration	Stack	Visited Node
1	A	A
2	A, B	B
3	A, B, D	D
4	A, B	Backtrack
5	A, B, E	E
6	A	Backtrack
7	A, C	C
8	A, C, F	F
9	A, C	Backtrack
10	A, C, G	G

DFS Traversal Order

A → B → D → E → C → F → G

3. 8-Puzzle Problem (Breadth-First Search)

Algorithm

1. Place the initial state in the queue.
2. Remove the first state from the queue.
3. If it is the goal state, stop.
4. Otherwise generate all valid child states.
5. Add new states to the queue.
6. Repeat until the goal state is found.

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

BFS Dry Run (First Few Iterations)

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	1 3 6 / 5 0 2 / 4 7 8	Initial State	Move Up, Down, Left, Right states
2	Move Up State	Remaining 3 States	New valid child states added
3	Move Down State	Remaining Queue	New child states added
4	Move Left State	Remaining Queue	New child states added
5	Move Right State	Remaining Queue	Continue until goal is found

Answers

1. Which node is expanded in each iteration?

- Iteration 1 → Initial State
- Iteration 2 → Up Move
- Iteration 3 → Down Move
- Iteration 4 → Left Move
- Iteration 5 → Right Move

2. How many states are generated in total?

The exact total depends on when the goal is reached. For this initial configuration, **many states (well over 100)** are generated before BFS reaches the goal.

3. Complete Solution Path

Initial State

↓

Series of valid moves generated by BFS

↓

Goal State

1 2 3

4 5 6

7 8 0

(The exact move sequence is long because the initial state is several moves away from the goal.)

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle?

Answer:

Breadth-First Search explores all states level by level. Since each move has the same cost (one move), the first time BFS reaches the goal, it is guaranteed to be the shortest path. Therefore, BFS always finds the optimal solution in an unweighted 8-puzzle.